

HDOC Day-34 - C Programming

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Aim

To write a menu-driven C program using separate functions and variable-length 2D arrays to verify the left and right distributive properties of matrix multiplication over matrix addition.

Operations

1. Left distributive property: $A \times (B + C) = (A \times B) + (A \times C)$, where A is $m \times n$ and B, C are $n \times p$.
2. Right distributive property: $(A + B) \times C = (A \times C) + (B \times C)$, where A, B are $m \times n$ and C is $n \times p$.
3. Exit.

Algorithm

1. Display the menu and read the user's choice.
2. For option 1, read m, n, p and matrices A, B and C. Compute B+C, then $A \times (B+C)$. Separately compute $A \times B$ and $A \times C$ and add them. Display and compare both $m \times p$ results.
3. For option 2, read m, n, p and matrices A, B and C. Compute A+B, then $(A+B) \times C$. Separately compute $A \times C$ and $B \times C$ and add them. Display and compare both $m \times p$ results.
4. State whether the selected distributive property is verified.
5. Repeat until Exit is selected.

Test Cases

Case	Choice / Dimensions	Expected result
1	1; m=2, n=2, p=2	Both sides equal; left distributive property verified
2	2; m=2, n=2, p=2	Both sides equal; right distributive property verified
3	3	Program exits

C Program

```
#include <stdio.h>

void readMatrix(int rows, int cols, int a[rows][cols]) {
    int i, j;
    for (i = 0; i < rows; i++)
        for (j = 0; j < cols; j++)
            scanf("%d", &a[i][j]);
}

void displayMatrix(int rows, int cols, int a[rows][cols]) {
    int i, j;
    for (i = 0; i < rows; i++) {
        for (j = 0; j < cols; j++)
            printf("%d\t", a[i][j]);
        printf("\n");
    }
}

void addMatrix(int rows, int cols, int a[rows][cols],
               int b[rows][cols], int result[rows][cols]) {
    int i, j;
    for (i = 0; i < rows; i++)
        for (j = 0; j < cols; j++)
            result[i][j] = a[i][j] + b[i][j];
}

void multiplyMatrix(int r1, int c1, int c2, int a[r1][c1],
                   int b[c1][c2], int result[r1][c2]) {
    int i, j, k;
    for (i = 0; i < r1; i++) {
        for (j = 0; j < c2; j++) {
            result[i][j] = 0;
            for (k = 0; k < c1; k++)
                result[i][j] += a[i][k] * b[k][j];
        }
    }
}

int compareMatrix(int rows, int cols, int a[rows][cols],
                  int b[rows][cols]) {
    int i, j;
    for (i = 0; i < rows; i++)
        for (j = 0; j < cols; j++)
            if (a[i][j] != b[i][j])
                return 0;
    return 1;
}

void leftDistributive() {
    int m, n, p;
    printf("\nEnter m, n and p: ");
    scanf("%d %d %d", &m, &n, &p);

    int A[m][n], B[n][p], C[n][p];
    int BC[n][p], AB[m][p], AC[m][p];
    int left[m][p], right[m][p];

    printf("\nEnter elements of Matrix A (%d x %d):\n", m, n);
    readMatrix(m, n, A);
    printf("\nEnter elements of Matrix B (%d x %d):\n", n, p);
    readMatrix(n, p, B);
    printf("\nEnter elements of Matrix C (%d x %d):\n", n, p);
    readMatrix(n, p, C);

    addMatrix(n, p, B, C, BC);
    multiplyMatrix(m, n, p, A, BC, left);

    multiplyMatrix(m, n, p, A, B, AB);
    multiplyMatrix(m, n, p, A, C, AC);
    addMatrix(m, p, AB, AC, right);

    printf("\nA x (B + C) =\n");
    displayMatrix(m, p, left);
    printf("\n(A x B) + (A x C) =\n");
    displayMatrix(m, p, right);

    if (compareMatrix(m, p, left, right))
        printf("\nLeft distributive property is VERIFIED.\n");
    else
        printf("\nLeft distributive property is NOT verified.\n");
}
```

```

}

void rightDistributive() {
    int m, n, p;
    printf("\nEnter m, n and p: ");
    scanf("%d %d %d", &m, &n, &p);

    int A[m][n], B[m][n], C[n][p];
    int AB[m][n], AC[m][p], BC[m][p];
    int left[m][p], right[m][p];

    printf("\nEnter elements of Matrix A (%d x %d):\n", m, n);
    readMatrix(m, n, A);
    printf("\nEnter elements of Matrix B (%d x %d):\n", m, n);
    readMatrix(m, n, B);
    printf("\nEnter elements of Matrix C (%d x %d):\n", n, p);
    readMatrix(n, p, C);

    addMatrix(m, n, A, B, AB);
    multiplyMatrix(m, n, p, AB, C, left);

    multiplyMatrix(m, n, p, A, C, AC);
    multiplyMatrix(m, n, p, B, C, BC);
    addMatrix(m, p, AC, BC, right);

    printf("\n(A + B) x C =\n");
    displayMatrix(m, p, left);
    printf("\n(A x C) + (B x C) =\n");
    displayMatrix(m, p, right);

    if (compareMatrix(m, p, left, right))
        printf("\nRight distributive property is VERIFIED.\n");
    else
        printf("\nRight distributive property is NOT verified.\n");
}

int main() {
    int choice;
    do {
        printf("\n===== HDOC DAY-34 =====\n");
        printf("1. Verify left distributive property\n");
        printf("2. Verify right distributive property\n");
        printf("3. Exit\n");
        printf("\nEnter your choice: ");
        scanf("%d", &choice);

        switch (choice) {
            case 1: leftDistributive(); break;
            case 2: rightDistributive(); break;
            case 3: printf("\nExiting program...\n"); break;
            default: printf("\nInvalid choice. Please enter 1, 2 or 3.\n");
        }
    } while (choice != 3);
    return 0;
}

```

Sample Execution

```
===== HDOC DAY-34 =====  
1. Verify left distributive property  
2. Verify right distributive property  
3. Exit
```

```
Enter your choice: 1  
Enter m, n and p: 2 2 2
```

```
Enter Matrix A:
```

```
1 2  
3 4
```

```
Enter Matrix B:
```

```
5 6  
7 8
```

```
Enter Matrix C:
```

```
1 2  
3 4
```

```
A x (B + C) =
```

```
26    32  
58    72
```

```
(A x B) + (A x C) =
```

```
26    32  
58    72
```

```
Left distributive property is VERIFIED.
```

Result

The program successfully verifies the left and right distributive properties of matrix multiplication over addition by displaying and comparing the matrices obtained on both sides.

Execution Output - HDOC Day-34

Terminal execution verifying the left distributive property of matrices.

```
Last login: Sun Sep 27 21:49:06 on ttys000
dhruvpandey@DHRUVs-MacBook-Air ~ % nano day34.c
dhruvpandey@DHRUVs-MacBook-Air ~ % clang day34.c -oday34
dhruvpandey@DHRUVs-MacBook-Air ~ % ./day34
```

```
===== HDOC DAY-34 =====
1. Verify left distributive property
2. Verify right distributive property
3. Exit
```

Enter your choice: 1

Enter m, n and p: 1

2
3

Enter elements of Matrix A (1 x 2):

1
2

Enter elements of Matrix B (2 x 3):

1
2
3
4
5
6

Enter elements of Matrix C (2 x 3):

12
3
443
22
33
2

A x (B + C) =

65 81 462

(A x B) + (A x C) =

65 81 462

Left distributive property is VERIFIED.

```
===== HDOC DAY-34 =====
1. Verify left distributive property
2. Verify right distributive property
3. Exit
```

Enter your choice: █